

KENDRIYA VIDYALAYA SANGATHAN, ERNAKULAM REGION

PRE BOARD EXAMINATION 2024-25

CLASS : X MATHEMATICS (BASIC)

Max. Marks: 80

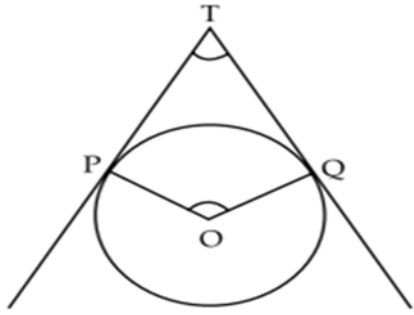
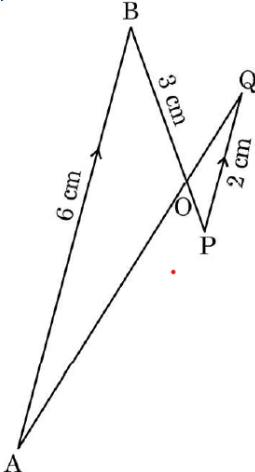
Duration: 3 Hours

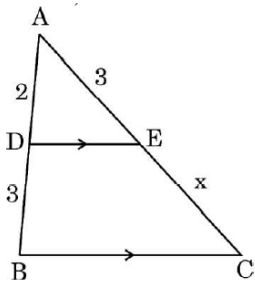
General Instructions:

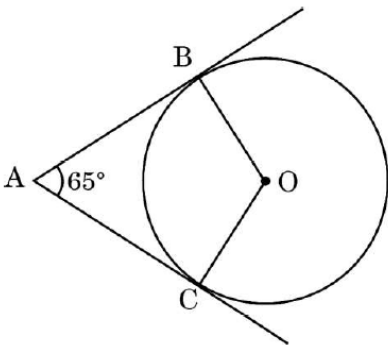
Read the following instructions carefully and follow them:

1. This question paper contains 38 questions.
2. This Question Paper is divided into 5 Sections A, B, C, D and E.
3. In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.
4. In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
5. In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.
6. In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.
7. In Section E, Questions no. 36-38 are case study based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
8. All Questions are compulsory. However, an internal choice in 2 Questions of section B, 2 Questions of section C and 2 Questions of section D has been provided. And internal choice has been provided in all the 2 marks questions of Section E.
9. Draw neat and clean figures wherever required.
10. Take $\pi = 22/7$ wherever required if not stated.
11. Use of calculators is not allowed.

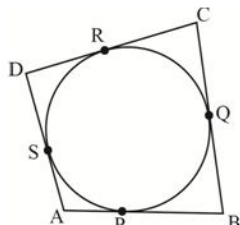
SECTION A		
Section A consists of 20 questions of 1 mark each.		
1	If $a = 2^3 \times 3$, $b = 2 \times 3 \times 5$, $c = 3^n \times 5$ and $\text{LCM}(a,b,c) = 2^3 \times 3^2 \times 5$, then n is (a) 1 (b) 2 (c) 3 (d) 4	1
2	The value of k for which the system of linear equations $x + 2y = 3$, $5x + ky + 7 = 0$ is inconsistent is (a) $-\frac{14}{3}$ (b) $\frac{2}{5}$ (c) 5 (d) 10	1
3	The quadratic equation $x^2 + 4x - 3\sqrt{2} = 0$ has (a) two distinct real roots (b) two equal real roots (c) no real roots (d) more than 2 real roots	1

4	If the distance between the points $(2, -2)$ and $(-1, x)$ is 5, one of the values of x is (a) -2 (b) 2 (c) -1 (d) 1	1
5	In Fig. TP and TQ are two tangents from an external point T to a circle with centre O. If $\angle POQ = 110^\circ$, then $\angle PTQ$ is:  (a) 60° (b) 70° (c) 80° (d) 90°	1
6	In the given figure, $AB \parallel PQ$. If $AB = 6$ cm, $PQ = 2$ cm and $OB = 3$ cm, then the length of OP is :  (a) 9 cm (b) 3 cm (c) 4 cm (d) 1 cm	1
7	If $\cos A = \frac{4}{5}$, then the value of $\tan A$ is (a) $\frac{4}{3}$ (b) $\frac{3}{5}$ (c) $\frac{3}{4}$ (d) $\frac{7}{5}$	1
8	In $\triangle ABC$ and $\triangle DEF$, $\frac{AB}{DE} = \frac{BC}{FD}$. Which of the following makes the two triangles similar? (a) $\angle A = \angle D$ (b) $\angle B = \angle D$ (c) $\angle B = \angle E$ (d) $\angle A = \angle F$	1
9	HCF of two numbers is 27 and their LCM is 162. If one of the numbers is 54, then the other number is (a) 36 (b) 35 (c) 9 (d) 81	1

10	The 21 st term of an AP whose first two terms are -3 and 4, is (a) 17 (b) 137 (c) 143 (d) -143	1
11	In the given figure, DE $\parallel$ BC. If AD = 2 units, DB = AE = 3 units and EC = x units, then the value of x is :  (a) 2 (b) 3 (c) 5 (d) $\frac{9}{2}$	1
12	From an external point Q, the length of tangent to a circle is 12 cm and the distance of Q from the centre of circle is 13 cm. The radius of circle (in cm) is (a) 10 (b) 5 (c) 12 (d) 7	1
13	The value of $2\sqrt{2}\cos 45^\circ \sin 30^\circ + 2\sqrt{3}\cos 30^\circ$ is (a) 1 (b) $\sqrt{3}$ (c) 4 (d) $\sqrt{2}$	1
14	If two solid hemispheres of same base radius r are joined together along their bases, then curved surface area of this new solid is (a) $4\pi r^2$ (b) $6\pi r^2$ (c) $3\pi r^2$ (d) $8\pi r^2$	1
15	An unbiased die is thrown. The probability of getting an odd prime number is : (a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) $\frac{2}{3}$	1
16	If $\frac{1}{2}$ is a root of the equation $x^2 + kx - \frac{5}{4} = 0$, then the value of k is (a) 2 (b) -2 (c) $\frac{1}{4}$ (d) $\frac{1}{2}$	1
17	The empirical relation between the mode, median and mean of a distribution is : (a) Mode = 3 Median – 2 Mean (b) Mode = 3 Mean – 2 Median (c) Mode = 2 Median – 3 Mean (d) Mode = 2 Mean – 3 Median	1
18	If α and β are the zeroes of the polynomial $2x^2 - 13x + 6$, then $\alpha + \beta$ is equal to (a) -3 (b) 3 (c) $\frac{13}{2}$ (d) $-\frac{13}{2}$	1

	DIRECTION: In the question number 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option A) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A) B) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A) C) Assertion (A) is true but reason (R) is false. D) Assertion (A) is false but reason (R) is true.	
19	Assertion (A) : Sum of first 10 terms of the arithmetic progression -0.5, -1.0, -1.5, is 31. Reason (R) : Sum of n terms of an AP is given as $S_n = \frac{n}{2} [2a + (n - 1)d]$ where a is first term and d common difference.	1
20	Assertion (A) : $\sqrt{2} + \sqrt{5}$ is an irrational number. Reason (R) : The sum of a rational number and an irrational number is an irrational number.	1
SECTION B		
Section B consists of 5 questions of 2 marks each.		
21	(a) Find a relation between x and y such that the point P(x, y) is equidistant from the points A (-5, 3) and B (7, 2). OR (b) Find the point on the x -axis which is equidistant from the points A(- 2, 3) and B(5, 4).	2
22	In the given figure, O is the centre of the circle. AB and AC are tangents drawn to the circle from point A. If $\angle BAC = 65^\circ$, then find the measure of $\angle BOC$. 	2

23	A student noted the number of cars passing through a spot on a road for 100 periods each of 3 minutes and summarized it in the table given below. Find the mode of the data: <table><tr><td>Number of cars</td><td>Frequency</td></tr><tr><td>0 - 10</td><td>7</td></tr><tr><td>10 - 20</td><td>14</td></tr><tr><td>20 - 30</td><td>13</td></tr><tr><td>30 - 40</td><td>12</td></tr><tr><td>40 - 50</td><td>20</td></tr><tr><td>50 - 60</td><td>11</td></tr><tr><td>60 - 70</td><td>15</td></tr><tr><td>70 - 80</td><td>8</td></tr></table>	Number of cars	Frequency	0 - 10	7	10 - 20	14	20 - 30	13	30 - 40	12	40 - 50	20	50 - 60	11	60 - 70	15	70 - 80	8	2
Number of cars	Frequency																			
0 - 10	7																			
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50 - 60	11																			
60 - 70	15																			
70 - 80	8																			
24	If $\tan (A + B) = \sqrt{3}$ and $\tan (A - B) = \frac{1}{\sqrt{3}}$; $0^{\circ} < A + B \leq 90^{\circ}$ $A > B$, find A and B.	2																		
25	(a) Determine the AP whose 3rd term is 5 and the 7th term is 9. OR (b) If the sum of the first 14 terms of an AP is 1050 and its first term is 10, find the 20th term.	2																		
SECTION C																				
Section C consists of 6 questions of 3 marks each.																				
26	Prove that $\sqrt{3}$ is an irrational number.	3																		
27	(a) Find the coordinates of the points of trisection of the line segment joining the points A(2, - 2) and B(- 7, 4). OR (b) If (1, 2), (4, y), (x, 6) and (3, 5) are the vertices of a parallelogram taken in order, find x and y.	3																		
28	Prove that $(\sin \theta + \operatorname{Cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = 7 + \tan^2 \theta + \cot^2 \theta$	3																		
29	The median of the distribution given below is 14.4. Find the value of x and y if the sum of the frequency is 20 <table><tr><td>Class Interval</td><td>Frequency</td></tr><tr><td>0 - 6</td><td>4</td></tr><tr><td>6 - 12</td><td>x</td></tr></table>	Class Interval	Frequency	0 - 6	4	6 - 12	x	3												
Class Interval	Frequency																			
0 - 6	4																			
6 - 12	x																			

		12 - 18	5		
		18 - 24	y		
		24 - 30	1		
30	(a) Prove that the lengths of tangents drawn from an external point to a circle are equal. OR (b) A quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$ 				3
31	If we add 1 to the numerator and subtract 1 from the denominator, a fraction reduces to 1. It becomes $\frac{1}{2}$ if we only add 1 to the denominator. What is the fraction?				3
SECTION D					
Section D consists of 4 questions of 5 marks each					
32	(a) A motor boat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream. OR (b) A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train.				5
33	State and prove basic proportionality theorem.				5
34	In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. Find: (i) the length of the arc (ii) area of the sector formed by the arc (iii) area of the segment formed by the corresponding chord.				5
35	(a) The angle of elevation of the top of a tower 30 m high from the foot of another tower in the same plane is 60° and the angle of elevation of the				5

	top of the second tower from the foot of the first tower is 30^0 . Find the distance between the two towers and also the height of the other tower. OR (b) From the top of a tower 100 m high, a man observes two cars on the opposite sides of the tower with angles of depression 30^0 and 45^0 respectively. Find the distance between the two cars.(Use $\sqrt{3} = 1.73$)													
SECTION E														
36	Computer-based learning (CBL) refers to any teaching methodology that makes use of computers for information transmission. At an elementary school level, computer applications can be used to display multimedia lesson plans. A survey was done on 1000 elementary and secondary schools of Assam and they were classified by the number of computers they had. <table border="1"><tr><td>Number of Computers</td><td>1 – 10</td><td>11 - 20</td><td>21 - 50</td><td>51 - 100</td><td>100 and more</td></tr><tr><td>Number of Schools</td><td>250</td><td>200</td><td>290</td><td>180</td><td>80</td></tr></table> One school is chosen at random. Then : (i) Find the probability that the school chosen at random has more than 100 computers. (ii) (a) Find the probability that the school chosen at random has 50 or fewer computers. OR (ii) (b) Find the probability that the school chosen at random has no more than 20 computers. (iii) Find the probability that the school chosen at random has 10 or less than 10 computers.	Number of Computers	1 – 10	11 - 20	21 - 50	51 - 100	100 and more	Number of Schools	250	200	290	180	80	1 2 2 1
Number of Computers	1 – 10	11 - 20	21 - 50	51 - 100	100 and more									
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37	Maximum Profit : An automobile manufacturer can produce up to 300 cars per day. The profit made from the sale of these vehicles can be modelled by the function $P(x) = -x^2 + 350x - 6600$ where $P(x)$ is the profit in thousand Rupees and x is the number of automobiles made and sold. Answer the following questions based on this model:													

	(i) What is the degree of the given polynomial? (ii) When no cars are produced what is a profit/loss (in thousand rupees)? (iii) (a) What is the break-even point ? (Zero profit point is called break even point) ? OR (iii) (b) What is the profit if 400 cars are produced ?	1 1 2 2
38	Silo : A silo is a structure for storing bulk materials. Silos are used in agriculture to store grain or fermented feed known as silage. Silos are commonly used for bulk storage of grain, coal, cement, carbon black, woodchips, food products and sawdust.  A silo is in the shape of cylinder surmounted by a conical top. The height and diameter of cylindrical part are 40 feet and 42 feet respectively and the slant height of conical part is 29 feet. (i) Find the height of the conical part. (ii) What is the ratio of the heights of the cylindrical part and conical part ? (iii) (a) How much metal sheet is required to make this silo ? OR (iii) (b) What is the storage capacity of silo?	1 1 2 2
